CARLO Y. MELONI

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RESEARCH INTERESTS

Language evolution; linguistic typology; historical linguistics; clause-linkage syntax; Semitic languages; Bayesian phylogenetics; computational and quantitative linguistics.

EMPLOYMENT HISTORY

Postdoctoral Researcher 2025 – 2027
University of Passau, Chair of Multilingual Computational Linguistics: Research on the *ProduSemy* project

Research Associate 2021 – 2025
Institute for the Interdisciplinary Study of Language Evolution: Research and teaching

Morphology and Syntax Expert 2021
Israel Innovation Authority: Morphological segmentation, POS-tagging and relational dependencies of a Hebrew online corpus using UD v2.8

Mandaic Transcription Expert 2017 – 2021
Department of Hebrew Language and Semitic Linguistics, Tel Aviv University: Transcriber and annotator of Mandaic texts for the Comprehensive Aramaic Lexicon project (PI: Prof. Matthew Morgenstern)

Research Assistant 2020
Linguistics Department, Tel Aviv University: Diachronic corpus searches and data coding for Prof. Mira Ariel

EDUCATION

Ph.D. in Linguistics 2021 – 2025
University of Zurich
Advisors: Prof. Balthasar Bickel, Dr. Chundra A. Cathcart, Prof. Jessica K. Ivani
"Syntactic Hierarchy in Linguistic Evolution: A Phylogenetic View of Clause-Linkage"
Final Grade: 6.0/6.0 (summa cum laude)

M.A. in Linguistics 2019 – 2021
Tel Aviv University
Advisors: Dr. Evan Gary Cohen, Prof. Hezy Mutzafi
Thesis Title: "The Reš Riddle: Identifying the Biblical Hebrew Rhotic"
Final Grade: 94/100 (magna cum laude)

B.A. in Linguistics (minor in Biology) 2016 – 2019
Tel Aviv University
Final Grade: 96/100 (summa cum laude)

PUBLICATIONS

Cathcart, Chundra, Arne Rubehn, Katja Bocklage, Luca Ciucci, Kellen Parker van Dam, Alžběta Kučerová, Jekaterina Mažara, **Meloni, Carlo Y.**, David Snee, and Johann-mattis List (2026). "How communicatively optimal are exact numeral systems? Once more on lexicon size and morphosyntactic complexity". In: *Proceedings of the Annual Meeting of the Cognitive Science Society*. Vol. 48. URL: https://escholarship.org/uc/item/4hz9j5f8.

Kučerová, Alvzběta, **Meloni, Carlo Y.**, and Johann-Mattis List (Aug. 2026). *Naming variation across languages is shaped by shared histories, semantics and familiarity*. DOI: 10.31234/osf.io/8tq2p_v1. URL: https://osf.io/preprints/psyarxiv/8tq2p_v1.

Meloni, Carlo Y., Chundra Cathcart, Erik J. Ringen, Jessica K. Ivani, and Balthasar Bickel (July 2025). "A Phylogenetic Perspective on the Syntax-Semantics Interface: A Case Study in Clause Linkage Constructions". In: *bioRxiv*. DOI: 10.1101/2025.07.19.665697. URL: https://www.biorxiv.org/content/early/2025/07/23/2025.07.19.665697.

Meloni, Carlo Y., Chundra A. Cathcart, Jessica K. Ivani, Taras Zakharko, Erik J. Ringen, and Balthasar Bickel (July 2025). "Linguistic evolution minimizes the amount of syntactic hierarchy in grammar". In: *bioRxiv*. DOI: 10.1101/2025.07.19.665693. URL: https://www.biorxiv.org/content/early/2025/07/23/2025.07.19.665693.

Meloni, Carlo Y. and Kellen Parker van Dam (2026). "A New Look at Avoidance Phenomena in Maltese Roots". In: *Journal of Maltese Linguistics*. Forthcoming.

Meloni, Carlo Y., Jessica K. Ivani, Taras Zakharko, Guanghao You, Chundra A. Cathcart, and Balthasar Bickel (July 2025). "Writing does not Impact the Evolution of Syntax". In: *bioRxiv*. DOI: 10.1101/2025.07.19.665695. URL: https://www.biorxiv.org/content/early/2025/07/23/2025.07.19.665695.

Meloni, Carlo Y. and Johann-Mattis List (Dec. 2025). "Towards a Unified Conversion Table for Semitic Transcriptions and Transliterations". In: *Computer-Assisted Language Comparison in Practice: Tutorials on Computational Approaches to the History and Diversity of Languages* 8.2, pp. 107–112. DOI: 10.15475/calcip.2025.2.6. URL: https://ojs.uni-passau.de/index.php/calcip/article/view/270.

Meloni, Carlo Y., Shauli Ravfogel, and Yoav Goldberg (June 2021). "Ab Antiquo: Neural Proto-language Reconstruction". In: *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Online: Association for Computational Linguistics, pp. 4460–4473. DOI: 10.18653/v1/2021.naacl-main.353. URL: https://aclanthology.org/2021.naacl-main.353.

INVITED TALKS & CONFERENCE PRESENTATIONS

A Quantitative Analysis of Biblical Hebrew's Rhotic 2026
Invited lecture at the European Hebrew Journal International Scholarly Conference

A Phylogenetic Perspective on the Syntax-Semantics Interface 2026
A paper presented at 16th International Conference of the Association for Linguistic Typology

Word Family Evolution in Semitic Languages 2026
A talk delivered at the workshop qc-leste: Quantitative Cross-Linguistic Explorations of Space, Time, and Evolution

Writing does not impact the evolutionary dynamics of syntax 2025
A poster presented at the 58th Annual Meeting of the Societas Linguistica Europaea

Writing does not impact the evolutionary dynamics of syntax 2025
A paper presented at the Emerging Topics in Typology at the Max Plank Institute for Evolutionary Anthropology

The Resh Riddle – Identifying the Biblical Hebrew Rhotic 2025

A poster presented at the 8th 'R-atics conference in Valletta

Writing does not impact the evolutionary dynamics of syntax 2025
A paper presented at the workshop "Bridging the gap: Unifying approaches to linguistic evolution" (University of Zurich)

Evolutionary dynamics of clause linkage constructions 2024
Invited lecture at the University of Passau colloquium

How dendrophilic is clause linkage? 2024
A poster presented at the 57th Annual Meeting of the Societas Linguistica Europaea

How dendrophilic is clause linkage? 2024
A paper presented at the IMPRS 2024 conference in Nijmegen

In Search of the Biblical Rhotic: Identifying Biblical Hebrew's Resh 2021
A poster presented at the 5th Edinburgh Symposium on Historical Phonology

L'histoire de la langue araméenne et son usage dans le différents cultes 2021
Invited lecture for the general public, Israelite Community of Lausanne and the Canton of Vaud

The Resh Riddle: Identifying the Biblical Hebrew Rhotic 2021
Second Temple Early Career Academy (STECA) summer conference

Ab Antiquo: Neural Proto-language Reconstruction 2021
A paper presented at the annual conference of the North American Chapter of the Association for Computational Linguistics (NAACL)

In Search for the Biblical Rhotic: A Phonological Reconstruction of Resh 2020
Tel Aviv University, Linguistics Department Colloquium

Ab Antiquo: Proto-language Reconstruction with RNNs 2019
A poster presented at the 57th meeting of the Association for Computational Linguistics (ACL), with Shauli Ravfogel and Yoav Goldberg

RNNs for proto-words reconstruction 2019
A paper presented at the 35th annual meeting of the Haiim B. Rosén Israeli Linguistic Society, with Shauli Ravfogel and Yoav Goldberg

AWARDS & GRANTS

Distributional Linguistics Lab Ph.D. Salary 2021 – 2025
Institute for the Interdisciplinary Study of Language Evolution, University of Zurich

Yohanan Goldman Scholarship for Outstanding MA students 2020
The Lester and Sally Entin Faculty of Humanities, Tel Aviv University

Travel grant from prof. Mira Ariel 2019
Linguistics Department, Tel Aviv University

Scholarship for Academic Excellence 2018
The Lester and Sally Entin Faculty of Humanities, Tel Aviv University

Dean's List 2018
The Lester and Sally Entin Faculty of Humanities, Tel Aviv University

Scholarship for Academic Excellence 2017
The Lester and Sally Entin Faculty of Humanities, Tel Aviv University

Dean's List 2017
The Lester and Sally Entin Faculty of Humanities, Tel Aviv University

Scholarship for undergraduate students 2016
Raffaele Cantoni Fellowship for Undergraduate students, The Hebrew University in Jerusalem

TEACHING (ASSISTANT ROLE)

Fieldwork Methods and Elicitation Techniques Spring 2024
1.5 hours/week, Department of Comparative Language Science, University of Zurich (MA class)

Methods in Evolutionary Linguistics Fall 2022, 2023
2 hours/week, Department of Comparative Language Science, University of Zurich (MA class)

Advanced Phonology Spring 2020
2 hours/week, Linguistics Department, Tel Aviv University (2nd year BA pro-seminar)

Intro to Linguists Fall 2019
2 hours/week, Linguistics Department, Tel Aviv University (1st year BA class)

ACADEMIC SERVICE

Reviewer
Linguistic Typology
Scientific Reports

TECHNICAL SKILLS

Programming:	R, Python, SQL
Statistical modelling:	Stan, Bayesian hierarchical modelling, Gaussian processes
Phylogenetics:	BEAST2, Bayesian phylogenetic comparative methods
Research computing:	Git, Linux, Slurm/HPC
Reproducible research:	Quarto, R Markdown, Pandoc, LaTeX

LANGUAGES

Italian	Native	Latin	Reading skills
Modern Hebrew	Native	Biblical Hebrew	Reading skills
English	Fluent	Mishnaic Hebrew	Reading skills
French	Proficient	Biblical Aramaic	Reading skills
Spanish	Proficient	Targumic Aramaic	Reading skills
Esperanto	Proficient	Jewish Babylonian Aramaic	Reading skills
German	Conversational	Syriac	Reading skills
Yiddish	Conversational	Mandaic	Reading skills
Russian	Conversational	Old Babylonian	Reading skills
Modern Standard Arabic	Passive	Ge'ez	Reading skills

ADDITIONAL TRAINING

Summer schools
- *The Gaussian Process and Uncertainty Quantification Summer School* 2026
  University of Manchester
- *Eighth Summer School on Statistical Methods for Linguistics and Psychology* 2024
  University of Potsdam

· *Bayesian phylogenetic inference with BEAST2* 2024
Transmitting Science
· *Seventh Summer School on Statistical Methods for Linguistics and Psychology* 2023
University of Potsdam
· *Mapping Trait Evolution* 2023
Transmitting Science
· *Lisbon Machine Learning School — LxMLS* 2022
Instituto Superior Técnico, Instituto de Telecomunicações, INESC-ID & Unbabel
· *Language universals and language diversity in an evolutionary perspective* 2019
Lake Como School of Advanced Studies

Scientific proficiency

· *Applied LLMs & NLP for Language Scientists* 2025
Institute for the Interdisciplinary Study of Language Evolution, Universität Zürich
· *Scientific Workflows on Science IT Infrastructure* 2025
Zentrale Informatik IT Fort- und Weiterbildungen, Universität Zürich
· *Introduction to the Linux command line* 2025
Zentrale Informatik IT Fort- und Weiterbildungen, Universität Zürich
· *Phython for Digital Humanities* 2025
Zentrale Informatik IT Fort- und Weiterbildungen, Universität Zürich
· *R: Crash Course in Statistics using R* 2023
Zentrale Informatik IT Fort- und Weiterbildungen, Universität Zürich
· *R: Reporting using R Markdown* 2022
Zentrale Informatik IT Fort- und Weiterbildungen, Universität Zürich
· *R: tidyverse for Data Science* 2022
Zentrale Informatik IT Fort- und Weiterbildungen, Universität Zürich
· *Introduction to Text Analytics with Python* 2022
Zentrale Informatik IT Fort- und Weiterbildungen, Universität Zürich
· *Intermediate Python* 2022
Zentrale Informatik IT Fort- und Weiterbildungen, Universität Zürich
· *Python – Data Analysis Essentials* 2022
Zentrale Informatik IT Fort- und Weiterbildungen, Universität Zürich
· *Introduction to Git and Gitlab* 2021
Zentrale Informatik IT Fort- und Weiterbildungen, Universität Zürich
· *Python Basics* 2021
Zentrale Informatik IT Fort- und Weiterbildungen, Universität Zürich
· LaTeX – *Introduction into the typesetting system* 2021
Zentrale Informatik IT Fort- und Weiterbildungen, Universität Zürich